Matrices as data and linear maps

This chapter works over the reals. In complex spaces, use conjugate transposes and Hermitian inner products, and replace coefficient squares by absolute squares. A grayscale image can be represented by $A \in \mathbb{R}^{m \times n}$, with pixel intensities as entries. Color images usually form tensors; a matrix treatment must specify how the channels are arranged.

The outer product has rank at most one; it has rank one exactly when both vectors are nonzero:

$$uv^T = \begin{bmatrix} u_1 \\ \vdots \\ u_m \end{bmatrix} \begin{bmatrix} v_1 & \cdots & v_n \end{bmatrix},$$

$$(uv^T)_{ij} = u_i v_j.$$

A rank-one factorization stores $m + n$ scalars; this is fewer than mn only when $m + n < mn$. SVD expresses a general matrix as a weighted sum of such pieces. Large singular values often capture substantial reconstruction energy.

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Full and compact SVD: check the dimensions first

Let A be $m \times n$, with rank r and $q=\min(m,n)$. Every real A admits $A=U\Sigma V^T$. The nonnegative singular values are ordered; exactly r are positive.

Form	U or U_r	Σ or Σ_r	V^T or V_r^T
Full	$m \times m$	$m \times n$	$n \times n$
Compact, used here	$m \times r$	$r \times r$	$r \times n$

Full U and V are orthogonal. Compact factors satisfy $U_r^T U_r = V_r^T V_r = I_r$, while $U_r U_r^T$ and $V_r V_r^T$ are projections, generally not identities. A thin interface may instead retain q directions including zero singular values. At $r=0$, $A=0$ is the empty sum.

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Right singular vectors from $A^T A$

$$x^T A^T A x = \|Ax\|^2 \geq 0.$$

Thus $A^T A$ is symmetric positive semidefinite and has spectral decomposition

$$A^T A = V \Lambda V^T.$$

Order the positive eigenvalues and set

$$\lambda_1 \geq \cdots \geq \lambda_r > 0,$$

$$\sigma_i = \sqrt{\lambda_i}.$$

For $i \leq r$,

$$\|Av_i\|^2 = v_i^T A^T A v_i = \sigma_i^2,$$

so $Av_i \neq 0$.

The vectors $v_1, \ldots, v_r$ form an orthonormal row-space basis. The vectors $v_{r+1}, \ldots, v_n$ form an orthonormal nullspace basis.

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Left singular vectors: normalize the nonzero outputs

For $i \le r$, set $u_i = Av_i/\sigma_i$. For any $i,j \le r$,

$$u_i^T u_j = \frac{v_i^T A^T A v_j}{\sigma_i \sigma_j} = \frac{\sigma_j^2 v_i^T v_j}{\sigma_i \sigma_j} = \delta_{ij}.$$

These orthonormal vectors lie in C(A); there are r of them, so they form a column-space basis. Complete them to an orthonormal basis U of $\mathbb{R}^m$ using vectors in $N(A^T)$. Do not divide by a zero singular value.

For an arbitrary $x = \sum c_j v_j$, the zero-eigenvalue directions have $Av_j = 0$. Therefore $Ax = \sum_{j=1}^{r} c_j \sigma_j u_j = U\Sigma V^T x$. Equality for every x proves the decomposition, including rectangular and rank-deficient matrices.

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Comparing $A^T A$ and AA^T

$$AA^T = U\Sigma\Sigma^T U^T.$$

The two Gram matrices have the same positive eigenvalues and rank. The multiplicity of zero can differ because their sizes can differ.

This explains the general structure.

- The eigenvectors of $A^T A$ give V.
- For positive singular values, choose $u_i = Av_i/\sigma_i$; independent arbitrary eigenbases of AA^T and $A^T A$ need not be paired, especially at repeated singular values.
- The positive eigenvalues are the squared singular values.
- Rectangular and rank-deficient matrices also admit SVD.

Numerical implementations usually avoid forming $A^T A$, which squares the relevant condition number and can overflow. Stable methods use bidiagonal reduction and algorithms for the resulting bidiagonal singular-value problem.

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Orthogonal outer products explain the squared-error sum

The compact SVD gives $A = \sum_{i=1}^{r} \sigma_i u_i v_i^T$. Define

$$\langle B, C \rangle_F = \operatorname{tr}(B^T C) = \sum_{ij} B_{ij} C_{ij}, \qquad \|B\|_F^2 = \sum_{ij} B_{ij}^2.$$

Since $\langle u_i v_i^T, u_j v_j^T \rangle_F = (u_i^T u_j)(v_i^T v_j) = \delta_{ij}$, the outer products form orthonormal matrix directions. Hence $\|A\|_F^2 = \sum_{i=1}^{r} \sigma_i^2$.

For $A \neq 0$, the retained squared-magnitude fraction is $\sum_{i=1}^{k} \sigma_i^2 / \sum_{i=1}^{r} \sigma_i^2$. It is not a guarantee of visual quality or downstream accuracy. The nuclear norm $\Sigma \sigma_i$ is a different quantity.

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Truncated SVD: retained directions and discarded errors

Let a real matrix A have rank r and decreasing positive singular values. Restrict approximations to rank at most k, initially with $1 \leq k < r$. Truncation retains the leading terms:

$$A_k = \sum_{i=1}^{k} \sigma_i u_i v_i^T, \qquad E = A - A_k = \sum_{i=k+1}^{r} \sigma_i u_i v_i^T.$$

Retained directions satisfy $A_k v_i = A v_i$. Discarded directions satisfy $A_k v_i = 0$, so $E v_i = \sigma_i u_i$. This constructs a candidate; it has not yet ruled out every other rank-k matrix.

Fix $A = \mathrm{diag}(9, 4, 1)$, k=1. Then $A_1 = \mathrm{diag}(9, 0, 0)$ and $E = \mathrm{diag}(0, 4, 1)$. First distinguish the two error measures, then prove their lower bounds. Here $u_i = v_i = e_i$; a general matrix can have different left and right directions.

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Two norms ask different questions about the same error

For $E = \operatorname{diag}(0, 4, 1)$:

Unit input	Error output	Squared length
e_1	0	0
e_2	$4e_2$	16
e_3	e_3	1

The spectral norm asks for the **largest error over all unit inputs**. If $\|x\| = 1$,

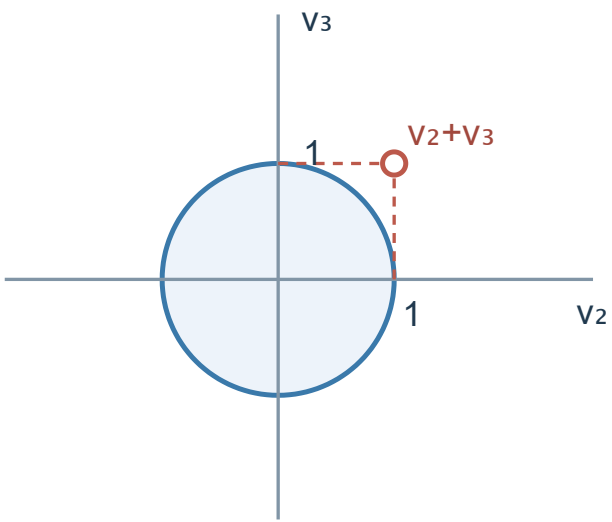
$$\|Ex\|^2 = 16x_2^2 + x_3^2 \le 16(x_1^2 + x_2^2 + x_3^2) = 16.$$

Input e_2 attains equality, hence $\|E\|_2 = 4$. The Frobenius norm instead sums squared column errors: $\|E\|_F^2 = 0 + 16 + 1 = 17$, so $\|E\|_F = \sqrt{17}$. It is not another maximum unit-input output length.

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Error ellipse: why $\sqrt{17}$ is not its longest semiaxis

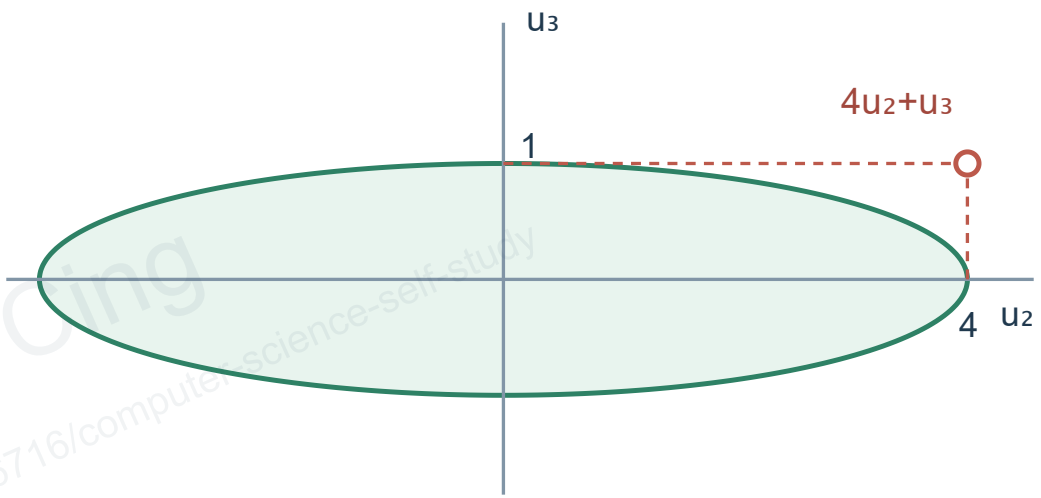
Input: v_2, v_3 (same unit scale)



$Ev_1 = 0$; red input length $\sqrt{2}$

E

Output: u_2, u_3 plane



Largest unit-input error = 4

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Error ellipse: why $\sqrt{17}$ is not its longest semiaxis (continued)

Since $Ev_1 = 0$, project the three-dimensional unit ball onto the v_2,v_3 unit disk, then apply E. Its image is the ellipse disk shown. The longest semiaxis is four. The red input $v_2 + v_3$ has length $\sqrt{2}$, not one. Its output length $\sqrt{17}$ therefore does not qualify for the spectral-norm maximum; normalizing the input gives output length $\sqrt{17/2} < 4$.

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Spectral lower bound: rank k cannot preserve k+1 dimensions

Take any B with rank at most k, without assuming shared singular vectors. On $S = \mathrm{span}\{v_1, \ldots, v_{k+1}\}$, B has rank at most k. Rank–nullity gives a unit $z \in S$ with $Bz = 0$.

Write $z = \sum_{i=1}^{k+1} c_i v_i$, where $\sum_i c_i^2 = 1$. Orthogonality gives

$$\|(A - B)z\|^2 = \|Az\|^2 = \sum_{i=1}^{k+1} \sigma_i^2 c_i^2 \geq \sigma_{k+1}^2.$$

Thus $\|A - B\|_2 \geq \sigma_{k+1}$. In our example, even the nondiagonal $B = \begin{bmatrix} 9 & 9 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ admits $z = (1, -1, 0)^T/\sqrt{2}$.

Then Bz=0 and $\|Az\|^2 = (81 + 16)/2 = 97/2 \geq 16$. The argument covers arbitrary candidates, not just diagonal ones.

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Frobenius lower bound: split each column at a right angle

Let P be the orthogonal projector onto $C(B)$. For each pair of columns a_j, b_j, $Pb_j = b_j$, so

$$a_j - b_j = (I - P)a_j + (Pa_j - b_j).$$

The first part is perpendicular to $C(B)$; the second lies in it. Apply Pythagoras column by column and sum:

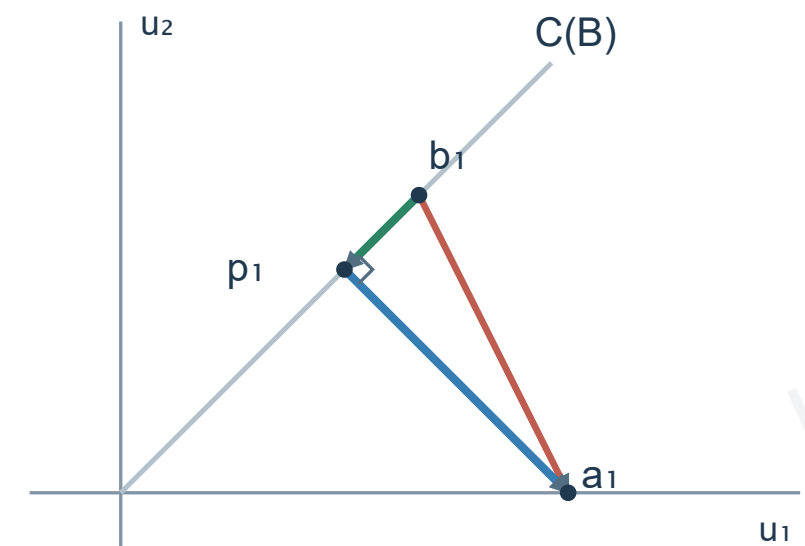
$$\|A - B\|_F^2 = \|(I - P)A\|_F^2 + \|PA - B\|_F^2 \geq \|(I - P)A\|_F^2.$$

Projection chooses the nearest point in the subspace. Choosing another point adds squared error. Complementary orthogonal projections also give

$$\|(I - P)A\|_F^2 = \|A\|_F^2 - \|PA\|_F^2 = \sum_{i=1}^{r} \sigma_i^2 - \operatorname{tr}(PAA^T).$$

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One projected column: 45 = 81/2 + 9/2



Blue: perpendicular error; green: in-subspace error

One column, in the first coordinate plane

$a_1=(9,0,0)$, $b_1=(6,6,0)$

$p_1=Pa_1=(9/2,9/2,0)$

$a_1-b_1 = (a_1-p_1) + (p_1-b_1)$

$a_1-p_1 \perp p_1-b_1$

$45 = 81/2 + 9/2$

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Use $C(B) = \text{span}\{(1,1,0)^T\}$, $a_1 = (9,0,0)^T$, $b_1 = (6,6,0)^T$. Projection gives $p_1 = (9/2, 9/2, 0)^T$. Error $(3,-6,0)^T$ splits as $(9/2,-9/2,0)^T + (-3/2,-3/2,0)^T$. Their dot product is zero and squared lengths give $45 = 81/2 + 9/2$. This depicts one column; Frobenius error sums all columns.

How much squared magnitude can a projection retain?

Let P project onto C(B). Complete the left singular basis and pad $\sigma_i=0$ for $i>r$. Set $\alpha_i=u_i^{\mathsf{T}}Pu_i=\|Pu_i\|^2$. Then $0\le\alpha_i\le1$ and $\sum\alpha_i=\operatorname{tr}P=\operatorname{rank}P\le k$.

Each direction has capacity at most 1; the total capacity is at most k. For $A=\operatorname{diag}(9,4,1)$, $k=1$, project onto span(1,1,0):

Direction	σ_i^2	α_i	Retained squared magnitude
$u_1=e_1$	81	1/2	81/2
$u_2=e_2$	16	1/2	8
$u_3=e_3$	1	0	0

The total is 97/2, below the 81 obtained by assigning all capacity to u_1. The next inequality proves the upper bound for every permitted projection, not only this example.

Prove the bound for retaining the largest k terms

Use $0 \le \alpha_i \le 1, \sum_i \alpha_i \le k$. Leading singular values are at least σ_k; all remaining ones are at most σ_k. Thus

$$\begin{aligned}\operatorname{tr}(PAA^T) &= \sum_{i=1}^{m} \sigma_i^2 \alpha_i \\ &\le \sum_{i=1}^{k} \sigma_i^2 \alpha_i + \sigma_k^2 \sum_{i=k+1}^{m} \alpha_i \\ &\le \sum_{i=1}^{k} \left[\sigma_i^2 \alpha_i + \sigma_k^2 (1 - \alpha_i)\right] \le \sum_{i=1}^{k} \sigma_i^2.\end{aligned}$$

Use tail values at most σ_k, total capacity at most k, and $\sigma_i^2 \ge \sigma_k^2$. Substitution yields $\|A - B\|_F^2 \ge \sum_{i=k+1}^{r} \sigma_i^2$.

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Eckart–Young: truncation attains both lower bounds

The preceding proofs cover every rank-at-most-k B. The truncated remainder E has exactly the discarded singular values, attaining both bounds:

$$\min_{\operatorname{rank} B \le k} \|A - B\|_2 = \sigma_{k+1}, \qquad \min_{\operatorname{rank} B \le k} \|A - B\|_F^2 = \sum_{i=k+1}^{r} \sigma_i^2.$$

For $A = \operatorname{diag}(9, 4, 1)$	Minimum spectral error	Minimum Frobenius error
k=0: only B=0	9	$\sqrt{98}$
k=1	4	$\sqrt{17}$
k=2	1	1
k≥3: B=A is allowed	0	0

The main proofs use $1 \le k < r$. At k=0 do not invoke σ_k: use the unique rank-zero matrix. At k≥r the error is zero. Appending a zero singular value leaves the earlier 9,4,1,0 example's k=1,2,3 errors unchanged.

Uniqueness depends on which norm you minimize

For A=diag(9,4,1), take $B_c = \mathrm{diag}(c, 0, 0)$. For every $5 \le c \le 13$, Bc has rank one and

$$\|A - B_c\|_2 = \max\{|9 - c|, 4, 1\} = 4.$$

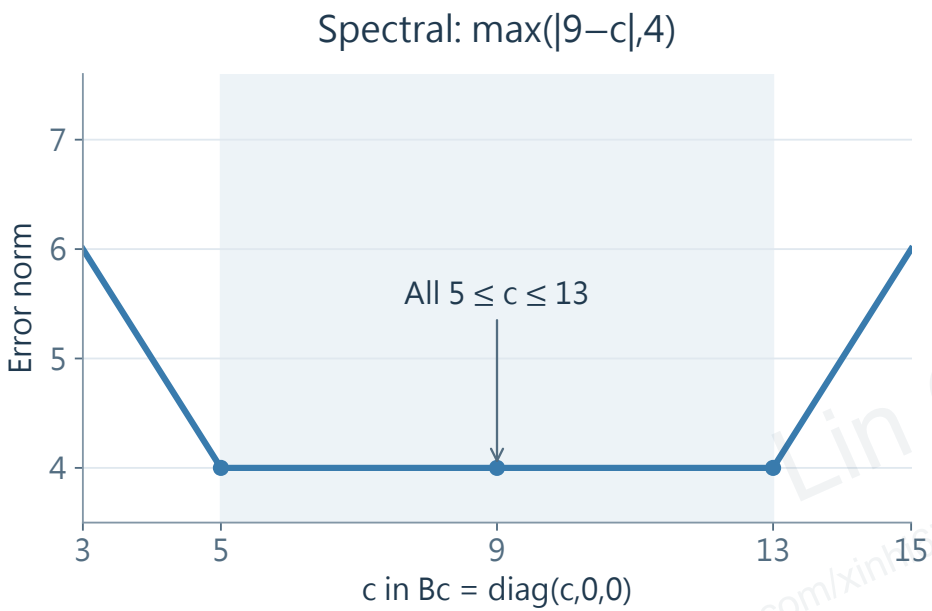
This whole family attains the global spectral lower bound despite the strict gap 9>4. In contrast,

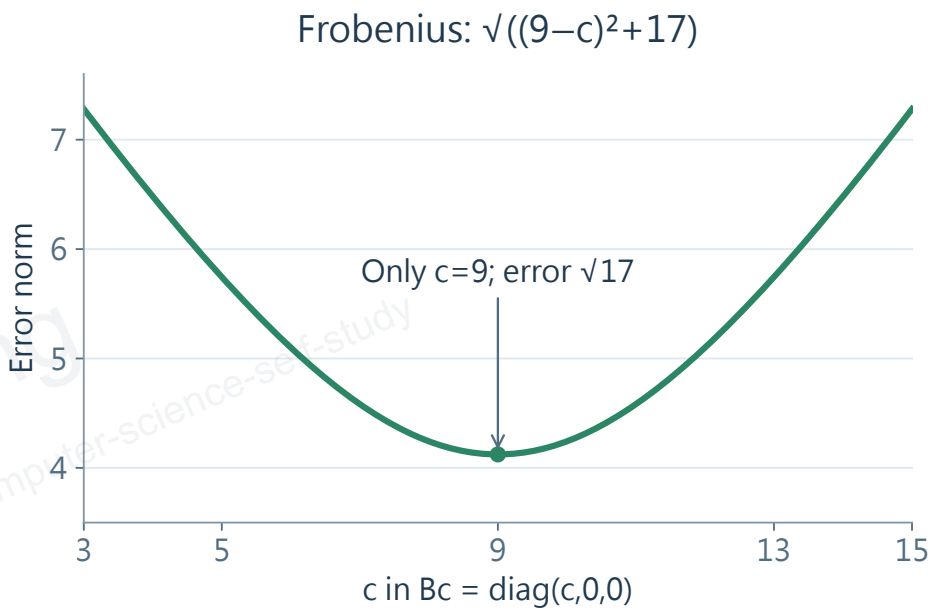
$$\|A - B_c\|_F^2 = (9 - c)^2 + 17$$

is minimized within this family only at c=9. More generally, for $1 \le k < r$, a strict gap $\sigma_k > \sigma_{k+1}$ makes the Frobenius minimizer unique. Equality in the trace bound forces the first k capacities to one and all others to zero; equality in Pythagoras forces B=PA, hence B=A_k. If the cutoff splits a positive tie, different retained parts of that fixed complete singular subspace can give different optimum matrices.

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The same candidates, plotted under two objectives





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The same candidates, plotted under two objectives (continued)

Both plots use the same Bc family. On the left, the discarded second direction already causes error four; changing the first direction has no effect on the maximum while its error stays at most four. On the right, its squared error still counts, so only c=9 minimizes the total. This family illustrates nonuniqueness without claiming to list every spectral minimizer.

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Storage cost of low-rank compression

Rank-k factors store

$$mk + nk + k = k(m + n + 1)$$

raw scalars, plus finite-precision metadata. Raw storage can decrease only if this count is below mn.

For a 1000×1000 image and rank 10, the factor count is 2.001×10^4, compared with 10^6 entries. Actual formats also store headers, quantization information, and other metadata.

Random full-rank data often compress poorly. Compressibility is a property of the data, not a guarantee supplied by SVD.

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The data orientation in PCA

Let

$$A_0 \in \mathbb{R}^{m \times n},$$

with variables in rows and samples in columns. With row means μ_j, center the data by

$$A_{js} = (A_0)_{js} - \mu_j.$$

The first principal direction u_1 maximizes the score vector's

$$s_1 = u_1^T A$$

sum of squares,

$$\|u_1^T A\|^2 = u_1^T A A^T u_1.$$

subject to $\|u_1\| = 1$. The spectral theorem gives maximum σ_1^2, attained at a dominant eigenvector of AA^T, the first left singular vector.

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Scores and sign ambiguity

From

$$A = U\Sigma V^T$$

the principal scores are

$$S = \Sigma V^T = U^T A.$$

The full S is m×n; compact scores $S_r = U_r^T A = \Sigma_r V_r^T$ have size r×n.

Row i lists all samples' scores on component i. Its squared norm equals σ_i^2, with zero rows included in a full representation.

Changing the signs of both u_i and v_i preserves the factorization. Changing only one changes the reconstruction. An individual sign has no intrinsic meaning.

Repeated singular values permit rotations of the component axes within the tied subspace. Compare subspace angles across PCA fits rather than matching arbitrary column numbers.

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Covariance denominators

For variables in rows and $n \geq 2$ samples, the usual sample covariance is

$$\widehat{C}_{n-1} = \frac{1}{n-1} A A^T,$$

The descriptive or population-denominator version is

$$\widehat{C}_n = \frac{1}{n} A A^T.$$

The eigenvectors coincide, while eigenvalues differ by factor $n/(n-1)$. For $n = 1$, the sample-variance denominator vanishes.

Unscaled PCA favors features with larger variance or larger numerical units. Standardization yields correlation-based PCA but changes the original geometry; it is a modeling choice.

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A two-dimensional PCA example

Samples:

$$(-1,-1), \quad (0,1), \quad (1,0).$$

Centered matrix:

$$A = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix}.$$

$$AA^T = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

$$u_1 = (1,1)^T/\sqrt{2}, \qquad u_2 = (1,-1)^T/\sqrt{2}.$$

Scores:

$$s_1 = \begin{bmatrix} -\sqrt{2} & 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix},$$

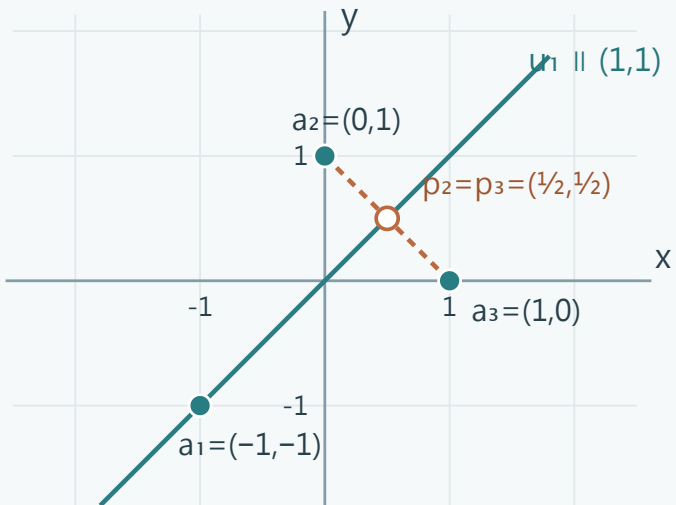
$$s_2 = \begin{bmatrix} 0 & -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}.$$

The score rows are orthogonal and their squared norms are 3 and 1.

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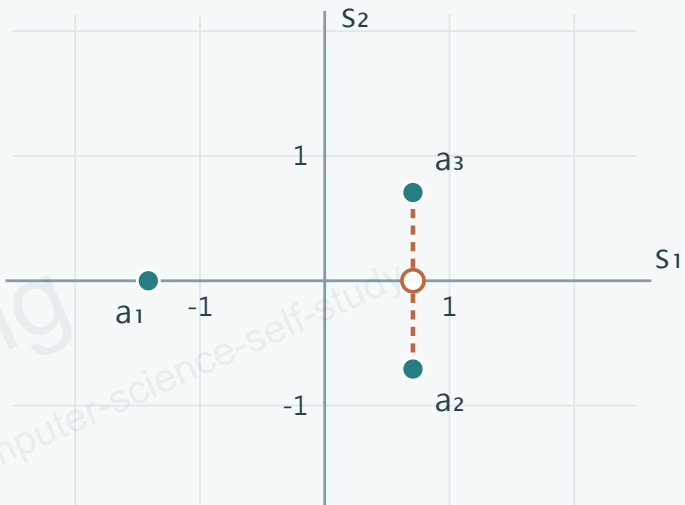
PCA projections and discarded perpendicular error

Original coordinates: project onto the first axis



Dashed: perpendicular residuals; open: shared projection

Score coordinates: the same points in a new basis



$s_1=(x+y)/\sqrt{2}$; $s_2=(x-y)/\sqrt{2}$

The first axis is y=x. Projecting the three samples gives (−1,−1), (1/2,1/2), (1/2,1/2). The last two projections coincide. Their residuals are (−1/2,1/2) and (1/2,−1/2), so the squared residual total is 1.

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PCA projections and discarded perpendicular error (continued)

The right panel uses $s_1 = u_1^T a$ and $s_2 = u_2^T a$ as coordinates. This is an orthogonal change of coordinates: point distances are preserved.

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Why maximum projected variance means minimum perpendicular error

For a unit direction u, split each centered sample into $uu^T a$ and $(I-uu^T)a$. Orthogonality gives

$$\sum_s \|(I-uu^T)a_s\|^2 = \|A\|_F^2 - \|u^T A\|^2.$$

The first term is fixed. Maximizing projected squared length therefore minimizes squared perpendicular distances to the line. Here the minimum is 4−3=1. For uncentered data, a best affine line is $\mu+\mathrm{span}(u_1)$.

The score rows are $(-\sqrt{2},1/\sqrt{2},1/\sqrt{2})$ and $(0,-1/\sqrt{2},1/\sqrt{2})$. Their squared lengths are 3 and 1; with sample denominator n−1=2, variances are 3/2 and 1/2, and the first explained fraction is 3/4. This differs from ordinary line fitting with vertical residuals and fixed horizontal coordinates.

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How singular vectors determine ellipse axes

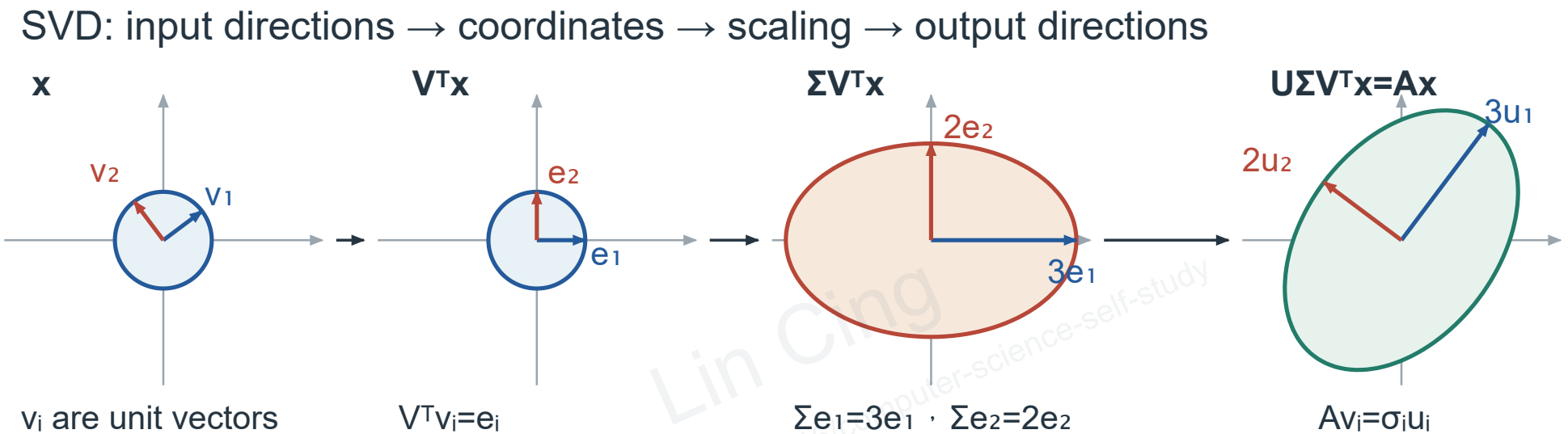
Take $V = \frac{1}{5}\begin{bmatrix} 4 & -3 \\ 3 & 4 \end{bmatrix}$, $U = \frac{1}{5}\begin{bmatrix} 3 & -4 \\ 4 & 3 \end{bmatrix}$, and Σ=diag(3,2). Both U and V are orthogonal, and

$$A = U\Sigma V^T = \frac{1}{5}\begin{bmatrix} 12 & -1 \\ 6 & 12 \end{bmatrix}.$$

First V^T sends v_i to e_i, then Σ stretches the axes, and U sets the output directions. A rotated circle looks unchanged as a set; the marked vectors still move.

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Equal scales: the semiaxis vectors are $\sigma_i u_i$

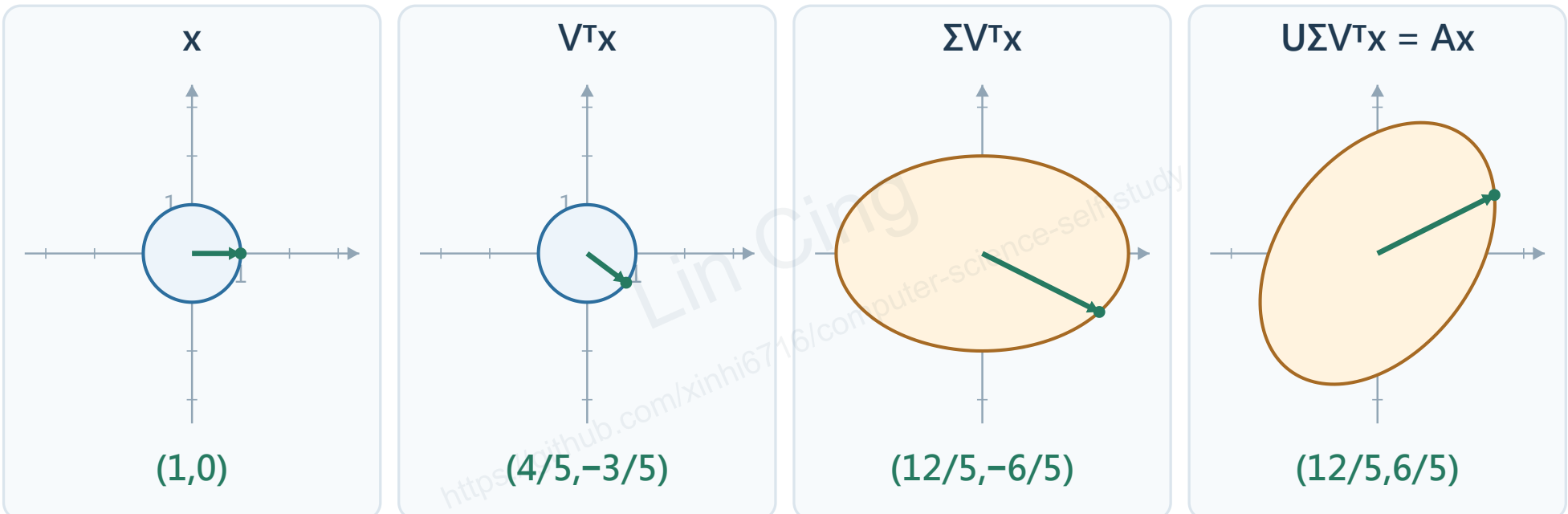


$Av_1=(9/5,12/5)^T=3u_1$ and $Av_2=(-8/5,6/5)^T=2u_2$. The semiaxis lengths are 3 and 2; unit u_i specify direction only. In this full-rank example the unit circle maps to the ellipse boundary and the disk to the filled ellipse.

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Track a unit vector outside the singular directions

Keep the preceding U, V and Σ, and take $x = (1, 0)^T$, which is neither singular direction. The green arrow tracks this point; each background curve is the image of the unit circle at that stage.



Same scale in every panel; the green vector is the same input tracked through the map.

Check coordinates and lengths separately

Stage	Resulting coordinates	Squared norm
Input x	(1,0)	1
$V^T x$	(4/5,−3/5)	1
$\Sigma V^T x$	(12/5,−6/5)	36/5
$U\Sigma V^T x = Ax$	(12/5,6/5)	36/5

V^T and U preserve lengths; Σ performs the scaling. Here $\|Ax\| = 6/\sqrt{5}$, which is neither singular value.

For any unit input, write $w = V^T x$. Then $w_1^2 + w_2^2 = 1$, and

$$\|Ax\|^2 = 9w_1^2 + 4w_2^2 = 4 + 5w_1^2 \in [4, 9].$$

Thus $2 \le \|Ax\| \le 3$, with the upper bound at $x = \pm v_1$ and lower bound at $x = \pm v_2$. This argument covers every unit input; one numerical trace alone would not prove the bounds.

SVD geometry with a right-angle rotation

Take $V = I$, $\Sigma = \operatorname{diag}(3, 2)$, and

$$U = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \qquad A = U\Sigma = \begin{bmatrix} 0 & -2 \\ 3 & 0 \end{bmatrix}.$$

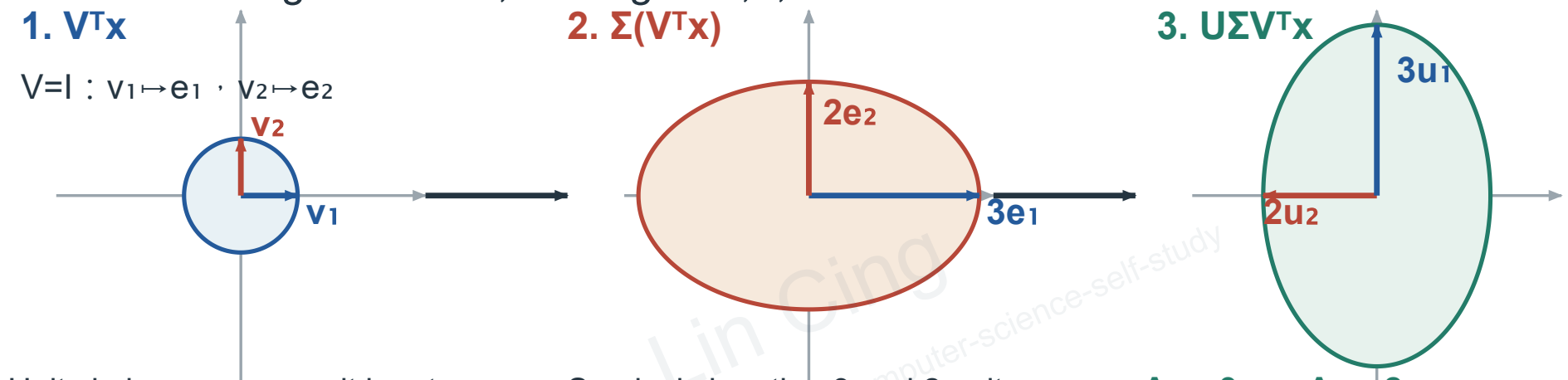
Stage	Action on coordinates	Geometric effect
V^T	(x_1, x_2)	Input directions unchanged
Σ	$(3x_1, 2x_2)$	Stretch to semiaxes 3 and 2
U	$(-2x_2, 3x_1)$	Rotate 90 degrees counterclockwise

Thus $Av_1 = (0, 3)^T = 3u_1$ and $Av_2 = (-2, 0)^T = 2u_2$. The unit disk maps to the filled ellipse; its boundary maps to the ellipse boundary. The next diagram tracks these same directions.

The maximum output length over unit inputs is $\sigma_1 = 3$; the minimum is $\sigma_2 = 2$. With a nontrivial nullspace, the unit sphere may map onto the filled image rather than only its surface.

Right-angle rotation: match each geometric stage

Three SVD stages: v axes, scaling σ=3,2, and u axes



Unit circle; v_1, v_2 are unit input axes

Semiaxis lengths: 3 and 2 units

$Av_1=3u_1$; $Av_2=2u_2$

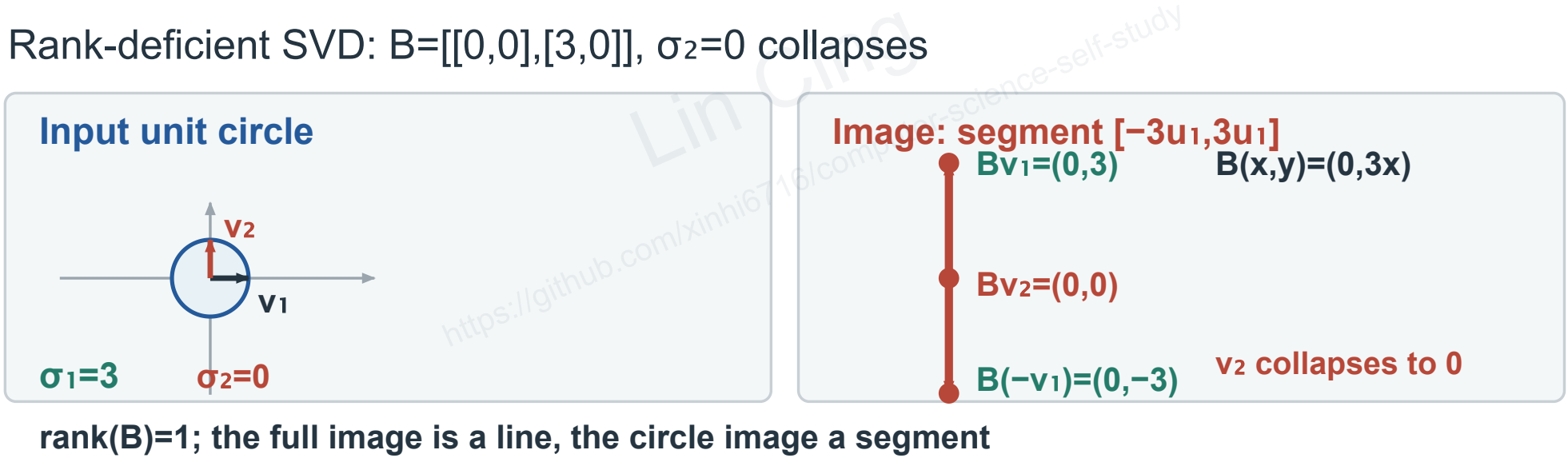
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A zero singular value collapses the disk to a segment

For deficient rank, take

$$B = \begin{bmatrix} 0 & 0 \\ 3 & 0 \end{bmatrix} = U \operatorname{diag}(3, 0) V^T.$$

Then $Bv_1 = (0, 3)^T$, while $Bv_2 = 0$. The v_2 direction collapses, and the unit disk maps onto the complete segment from $(0, -3)^T$ to $(0, 3)^T$.



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Spectral norm and condition number

$$\|A\|_2 = \max_{x \neq 0} \frac{\|Ax\|_2}{\|x\|_2}.$$

Expand

$$x = \sum_i c_i v_i,$$

$$\|Ax\|^2 = \sum_i \sigma_i^2 |c_i|^2 \leq \sigma_1^2 \|x\|^2.$$

Equality holds at $x = v_1$, so

$$\|A\|_2 = \sigma_1.$$

For full column rank, the usual singular-value condition number is

$$\kappa_2(A) = \frac{\sigma_{\max}}{\sigma_{\min}}.$$

An orthogonal square matrix has every singular value equal to 1 and condition number 1. A rank-deficient matrix has a zero smallest singular value and no finite ordinary inverse condition number.

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Distance to singularity and numerical rank

For an invertible square $A \in \mathbb{R}^{n \times n}$, remove $\sigma_n u_n v_n^T$.

$$\|A - A_0\|_2 = \sigma_n.$$

Thus σ_n is the spectral-norm distance to singularity. More generally, A_k is a nearest rank-at-most-k matrix, at distance σ_{k+1}.

Numerical rank uses a tolerance τ.

$$\sigma_i > \tau \|A\|.$$

The tolerance must reflect noise, dimensions, and downstream sensitivity. Stable singular values do not imply stable individual singular vectors, especially near repeated or closely spaced values.

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Perturbations bound each singular value

For equally sized $m \times n$ matrices A, B, let $q = \min(m, n)$, $1 \le i \le q$, pad missing singular values with zeros, and set $\delta = \|A - B\|_2$. Eckart–Young gives $\sigma_i(A) = \min_{\operatorname{rank} C < i} \|A - C\|_2$. Choosing $C = B_{i-1}$ yields

$$\sigma_i(A) \le \|A - B_{i-1}\|_2 \le \delta + \|B - B_{i-1}\|_2 = \delta + \sigma_i(B).$$

Exchanging A, B proves $|\sigma_i(A) - \sigma_i(B)| \le \delta$. This absolute bound does not guarantee a small relative error for tiny singular values.

Set $r_\delta = \#\{i : \sigma_i(A) > \delta\}$. Every compatible true matrix B has rank at least r_δ, since these singular values remain positive. Conversely, truncation to A_{r_δ} has error at most δ, so such a rank is attainable. If $r_\delta = q$, use $A_q = A$.

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A noise bound gives a minimum compatible rank

Let $A = \operatorname{diag}(1, 10^{-8}, 10^{-12})$ and $\delta = 10^{-10}$.

Compatible model	Rank	Distance from observed A
$\operatorname{diag}(1, 10^{-8}, 0)$	2	$10^{-12} \le \delta$
A itself	3	0

The first two singular values exceed the noise bound, so rank below 2 is impossible. Both rows are compatible; the data do not determine whether the third direction is exactly zero.

For a fixed absolute threshold t, a classification is robust if $\sigma_i(A) > t + \delta$ or $\sigma_i(A) < t - \delta$. Values within this band are uncertain. A relative threshold $t = \tau \|A\|_2$ also needs a recorded scale and noise model.

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QR pivots are not singular values

In exact arithmetic, $A = Q_1 R$, $Q_1^T Q_1 = I$, gives $A^T A = R^T R$, so A and R share singular values. However, the diagonal of R is not its singular-value list. Take

$$R = \begin{bmatrix} 1 & K \\ 0 & 1 \end{bmatrix}, \qquad K \geq 1.$$

Both diagonal entries are 1, but $\|Re_2\|_2 = \sqrt{K^2 + 1}$, giving $\sigma_{\max} \geq K$. Since $\sigma_{\max}\sigma_{\min} = |\det R| = 1$, we obtain $\sigma_{\min} \leq 1/K$ and $\kappa_2(R) \geq K^2$. A large off-diagonal entry can conceal near dependence from the diagonal alone.

The sizes of R_{ii} can be a heuristic, not a singular-value perturbation theorem. Column-pivoted QR often estimates numerical rank, but guaranteed accuracy requires explicit rank-revealing bounds or an SVD check. A nonzero floating-point pivot need not represent a trustworthy data direction.

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Polar decomposition

$$A = U\Sigma V^T = (UV^T)(V\Sigma V^T).$$

This page concerns square $A \in \mathbb{R}^{n\times n}$, so U, V, Σ all have size $n \times n$. Set

$$Q = UV^T, \qquad S = V\Sigma V^T.$$

Then Q is orthogonal and S symmetric positive semidefinite, giving

$$A = QS.$$

If A is invertible, every singular value is positive, so S is positive definite. The sign of $\det Q$ records orientation reversal. The factors separate an orthogonal rotation or reflection from a symmetric stretch.

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The pseudoinverse and its two projections

For compact $A = U_r \Sigma_r V_r^T$, invert only positive singular values:

$$A^+ = V_r \Sigma_r^{-1} U_r^T, \qquad AA^+ = U_r U_r^T, \quad A^+ A = V_r V_r^T.$$

These are orthogonal projections onto column space and row space. Orthonormal columns prove symmetry and idempotence; multiplying gives $AA^+A = A$ and $A^+AA^+ = A^+$. These are the four Moore–Penrose conditions.

Assumption	Equivalent formula
Full column rank	$A^+ = (A^T A)^{-1} A^T$
Full row rank	$A^+ = A^T (AA^T)^{-1}$

Without the stated rank assumption, the displayed inverse need not exist. The compact formula also handles rank zero as an empty product, giving $A^+ = 0$.

Transposing a rectangular matrix swaps singular directions

Take $B = \begin{bmatrix} -1 & 0 \\ 1 & -1 \\ 0 & 1 \end{bmatrix}$, and set $A = B^T$. Then

$$B^T B = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}, \qquad \sigma_1 = \sqrt{3}, \quad \sigma_2 = 1.$$

Choose $v_1 = (1, -1)^T/\sqrt{2}$, $v_2 = (1, 1)^T/\sqrt{2}$. Using $u_i = Bv_i/\sigma_i$ gives

$$u_1 = (-1, 2, -1)^T/\sqrt{6}, \quad u_2 = (-1, 0, 1)^T/\sqrt{2}.$$

Complete with $u_3 = (1, 1, 1)^T/\sqrt{3} \in N(B^T)$. If $B = U\Sigma V^T$, then $A = V\Sigma^T U^T$: the factors and their dimensions swap, as do the associated zero directions.

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Transposition changes the least-squares problem

For $Bx = b$, $b = (1, 0, 0)^T$, full column rank and the normal equations give the unique least-squares solution $\hat{x} = (-2/3, -1/3)^T$. Its projection and residual are

$$B\hat{x} = (2/3, -1/3, -1/3)^T, \quad e = b - B\hat{x} = (1, 1, 1)^T/3.$$

Here $B^T e = 0$ and $\|e\| = 1/\sqrt{3}$, so no exact solution exists.

Now set $A = B^T$ and $c = (1, 0)^T$. The equation $Ax = c$ is consistent. Its unique minimum-norm solution is $x^+ = (-2/3, 1/3, 1/3)^T$; all solutions are $x^+ + t(1, 1, 1)^T$. Since x^+ is orthogonal to the nullspace, adding a nonzero nullspace component increases the norm.

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A product of pseudoinverses need not reverse

AB = [1]

A = [1 0] · B = [1; 1]

A: 1×2 B: 2×1

B^+A^+ = [½]

B^+ = [½ ½] · A^+ = [1; 0]

B^+: 1×2 A^+: 2×1

$(AB)^+ = 1 \neq ½ = B^+A^+$

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Here $A = [1\ 0]$ has full row rank and $B = [1\ 1]^T$ has full column rank. Both attain their maximum possible rank, but $(AB)^+ = 1 \neq 1/2 = B^+A^+$. For invertible square factors the familiar inverse rule applies; rectangular pseudoinverses require additional conditions.

Why the pseudoinverse gives minimum norm

Expand $x = \sum_{i=1}^{n} c_i v_i$ in the full right singular basis. Orthogonality separates the residual:

$$\|b - Ax\|^2 = \sum_{i=1}^{r} (u_i^T b - \sigma_i c_i)^2 + \sum_{i=r+1}^{m} (u_i^T b)^2.$$

The second sum cannot be changed by x. Minimize each term in the first sum by taking $c_i = u_i^T b / \sigma_i$. Coefficients for $i > r$ are free because they lie in the nullspace. Hence every least-squares solution is

$$x = A^+ b + z, \quad z \in N(A), \qquad \|x\|^2 = \|A^+ b\|^2 + \|z\|^2.$$

The two terms are orthogonal because $A^+ b$ lies in row space. The unique minimum-norm choice is $z = 0$, even when the least-squares solution itself is nonunique.

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Rank-deficient least squares: one constraint, many solutions

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} = \sqrt{10}\, u_1 v_1^T, \quad u_1 = \frac{(1,2)^T}{\sqrt{5}}, \quad v_1 = \frac{(1,1)^T}{\sqrt{2}}.$$

The left vector describes an output direction; the right vector describes an input direction. Rank one does not make these vectors equal.

For $b = (1,3)^T$, the normal equations are

$$\begin{bmatrix} 5 & 5 \\ 5 & 5 \end{bmatrix} x = \begin{bmatrix} 7 \\ 7 \end{bmatrix}.$$

Both rows impose the same constraint $x_1 + x_2 = 7/5$. Every point on this line is a least-squares solution. The singular normal matrix cannot be inverted; the next calculation selects the point on this line closest to the origin.

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The minimum-norm solution

$$\sigma_1 u_1^T b = \sqrt{10}\frac{1+6}{\sqrt{5}} = 7\sqrt{2}.$$

$$x^+ = A^+ b = \frac{1}{\sigma_1} v_1 u_1^T b = \frac{7}{10}(1,1)^T.$$

Projection:

$$AA^+ b = u_1 u_1^T b = \frac{7}{5}(1,2)^T.$$

Residual:

$$e = b - AA^+ b = \left(-\frac{2}{5}, \frac{1}{5}\right)^T.$$

$$e^T(1,2) = 0,$$

The residual is therefore orthogonal to column space.

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Numerical workflow

SVD workflow:

1. Check shape, missing values, and units.
2. Center PCA data and standardize when the modeling choice requires it.
3. Use a stable SVD algorithm without forming normal equations.
4. Inspect singular-value gaps and numerical rank.
5. Choose k from the error budget.
6. Validate on held-out data or the downstream task.

Signs can flip and tied axes can rotate. Linear reconstruction alone establishes neither causal importance nor fairness.

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The four subspaces

Space	Ambient space	SVD basis	Dimension
row space	$\mathbb{R}^n$	$v_1, \ldots, v_r$	r
nullspace	$\mathbb{R}^n$	$v_{r+1}, \ldots, v_n$	$n - r$
column space	$\mathbb{R}^m$	$u_1, \ldots, u_r$	r
left nullspace	$\mathbb{R}^m$	$u_{r+1}, \ldots, u_m$	$m - r$

Input and output singular vectors belong to different roles and cannot generally be paired by an inner product. Equal dimensions allow an explicit coordinate identification, but do not make the bases identical. SVD pairs them through

$$Av_i = \sigma_i u_i.$$

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Chapter review

- SVD pairs left and right singular directions; never divide by a zero singular value.
- Truncated SVD error and uniqueness depend on the norm.
- PCA requires explicit data orientation, centering, and covariance denominator.
- The pseudoinverse first minimizes residual, then selects the minimum-norm solution.

References and attribution

Lin Cing, handwritten linear algebra notes. Gilbert Strang, Introduction to Linear Algebra, 5th ed., Chapter 7.

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